Data-driven SPS/DPS separation in J/ψ-pair production

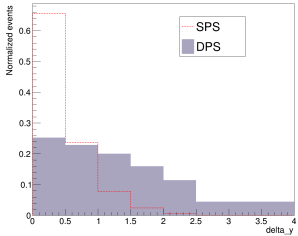
Hongbo Liao[1], Jinfeng Liu[2], Kai Yi[3], Lekai Yao[2], Muhammad Ahmad[2,4], Ruobing Tu[1], Shiyang Chen[2], Shunliang Zhang[2], Taozhe Yu[1], and Zhen Hu[2]

29 August 2026

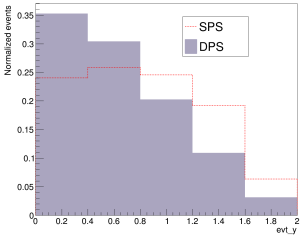
[1]IHEP [2]Tsinghua University [3]Nanjing Normal University
[4]Texas A&M University

AN template fit: separating SPS and DPS shapes

MC samples provide normalized SPS and DPS shapes in two discriminating observables.



$|\Delta y(\mathrm{J}/\psi, \mathrm{J}/\psi)|$: DPS extends to larger separation



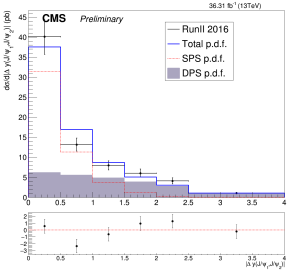
$|y(\mathrm{J}/\psi \mathrm{J}/\psi)|$: complementary SPS/DPS shape information

One shared DPS fraction in the simultaneous fit

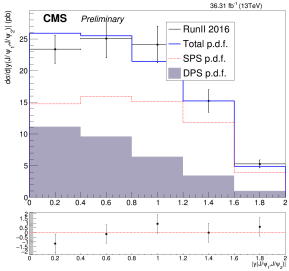
$$p_i^{\mathrm{fit}} = (1 - f_{\mathrm{DPS}})\, T_i^{\mathrm{SPS}} + f_{\mathrm{DPS}}\, T_i^{\mathrm{DPS}}$$

AN simultaneous template-fit result

The same f_{DPS} is fitted to both differential cross sections.



$\mathrm{d}\sigma/\mathrm{d}|\Delta y|$



$\mathrm{d}\sigma/\mathrm{d}|y(\mathrm{J}/\psi\mathrm{J}/\psi)|$

Extracted DPS fraction

$$f_{\mathrm{DPS}} = 0.34 \pm 0.10\ (\mathrm{stat.})^{+0.02}_{-0.17}\ (\mathrm{syst.})$$

Systematic uncertainty from individual–simultaneous fit differences

Why pursue a data-driven separation?

The two approaches separate the same SPS/DPS components with different inputs.

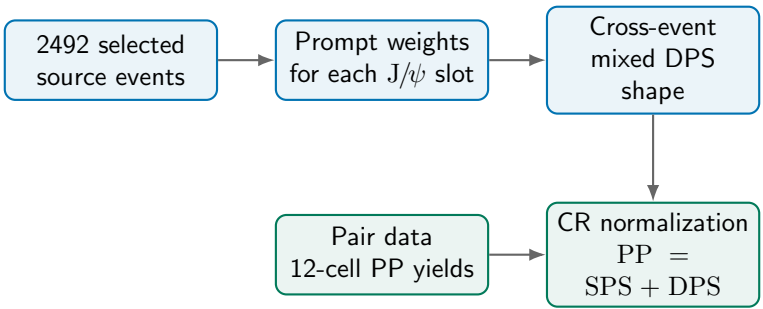
	MC template fit	Data-driven separation
Primary input	Simulated SPS and DPS shapes	Observed single-J/ψ and pair data
Main limitation	**Generator dependence**, especially the SPS shape	Mixing, factorization and control-region choices
Main advantage	Fits measured differential cross sections directly	Builds templates from data kinematics and occupancy

Goal: test the SPS/DPS separation with substantially different inputs and assumptions.

Data-driven workflow: from source events to SPS/DPS

Shape information
prompt-weighted single-J/ψ event mixing

Normalization information
fitted prompt-prompt yields in pair data

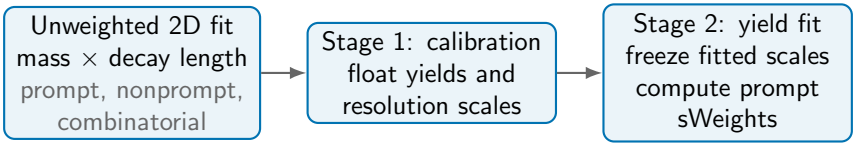


"Each J/ψ slot" means independent single-J/ψ fits for J/ψ_1 and J/ψ_2.

The separation is built from two data constraints with different roles.

Single-J/ψ source: isolate prompt candidates before mixing

Fit the two J/ψ slots separately; no transverse-momentum ordering is assumed.

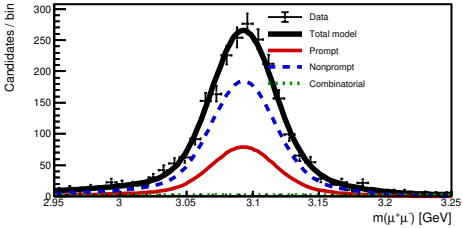


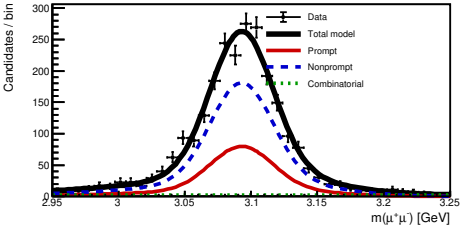
Slot	Fit entries	Prompt yield	$\sum w_{\mathrm{prompt}}$
J/ψ_1	2473	716.92	716.92
J/ψ_2	2474	728.72	728.72

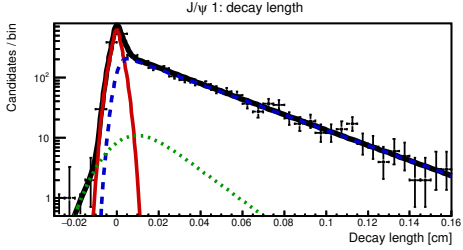
Fit acceptance
status $= 0$, covQual ≥ 2, EDM < 0.01

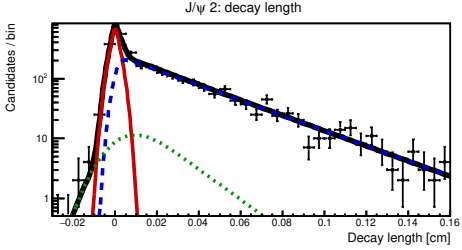
Signed prompt sWeights are retained

Prompt single-J/ψ extraction in data





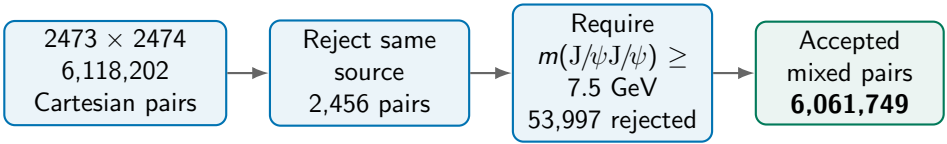




The two J/ψ slots are fitted independently; prompt sWeights feed event mixing.

Deterministic cross-slot all-pairs mixing

Build every allowed $\mathrm{J}/\psi_1(A) \otimes \mathrm{J}/\psi_2(B)$ pair with source events $A \neq B$.



Pair weight

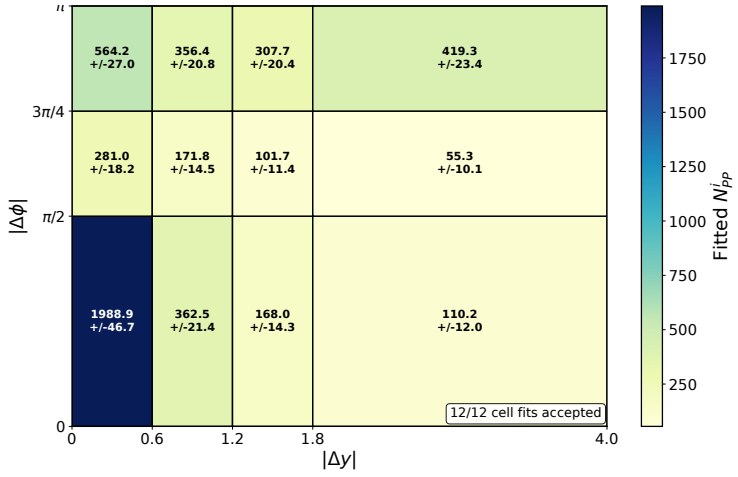
$$w_{AB} = w_{1,A}^{\mathrm{prompt}}\, w_{2,B}^{\mathrm{prompt}}\, C_{\mathrm{pair}}$$

- all pairs; no random sampling or mixing seed
- same source identified before pairing
- add the two J/ψ four-vectors; require $m(\mathrm{J}/\psi\mathrm{J}/\psi) \geq 7.5$ GeV
- positive and negative weighted pairs retained
- then apply C_{pair}: acceptance/efficiency weight

Method assumption: candidates from different source events represent independent scatters.

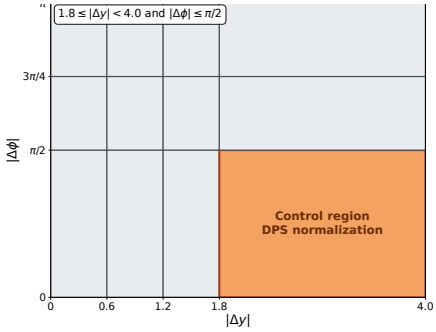
Prompt–prompt yields in the common phase space

A four-dimensional pair fit provides N^{i}_{PP} in each $(|\Delta y|, |\Delta\phi|)$ cell.



All 12 cell fits are accepted; the fitted uncertainty is shown in every cell.

The control region fixes the DPS normalization



Normalize the mixed template

$$\alpha = \frac{N_{\mathrm{PP}}^{\mathrm{CR}}}{N_{\mathrm{mix}}^{\mathrm{CR}}},$$
$$N_{\mathrm{DPS}}^{i} = \alpha N_{\mathrm{mix}}^{i}$$

Yield definitions

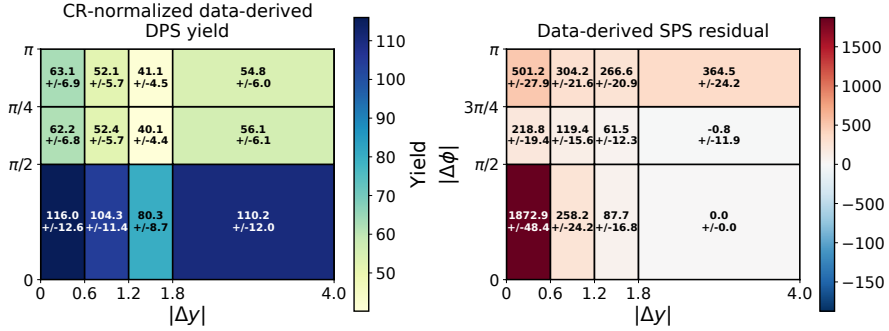
$N_{\mathrm{PP}}^{\mathrm{CR}}$: fitted prompt–prompt yield in the CR
$N_{\mathrm{mix}}^{\mathrm{CR}}$, N_{mix}^{i}: mixed-event yield in the CR / cell i
N_{DPS}^{i}: normalized DPS estimate in cell i

Normalization assumption

Highlighted cell treated as DPS-only; SPS leakage set to zero.

CR equality is imposed by normalization; it is not a purity measurement.

Cell-by-cell SPS/DPS decomposition



$$N^i_{\mathrm{SPS}} = N^i_{\mathrm{PP}} - N^i_{\mathrm{DPS}}$$

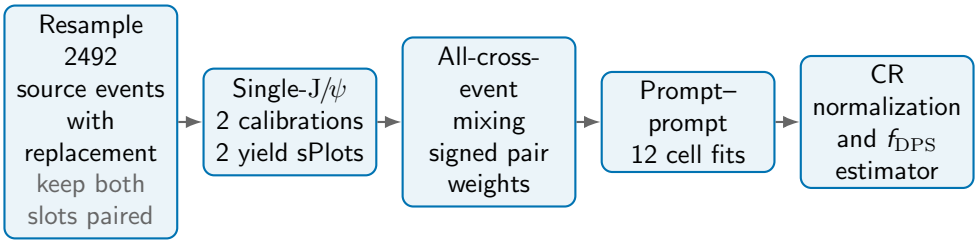
Signed residuals are retained; no cell is clipped.

Integrated DPS fraction

$$f_{\mathrm{DPS}} = \frac{832.86}{4887.01} = \mathbf{0.17042}$$

Full source-event bootstrap reruns the complete estimator

The resampling unit is the selected source event, not an individual mixed pair.

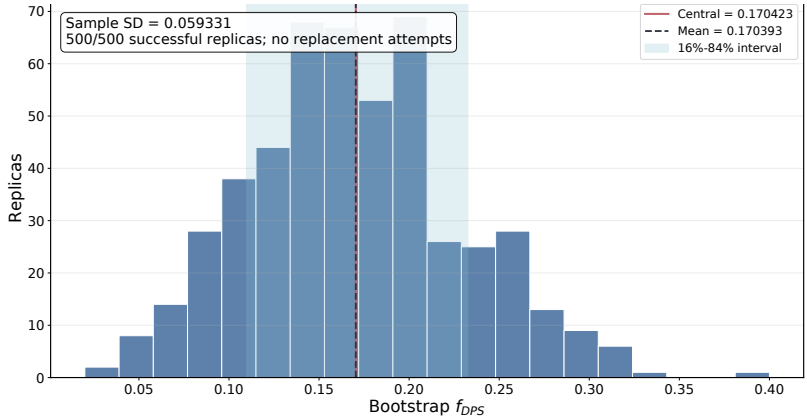


One bootstrap replica

One resampled source dataset $\longrightarrow$ one complete f_{DPS} extraction

The replica spread propagates fluctuations and correlations through the full chain.

Bootstrap ensemble gives the statistical uncertainty

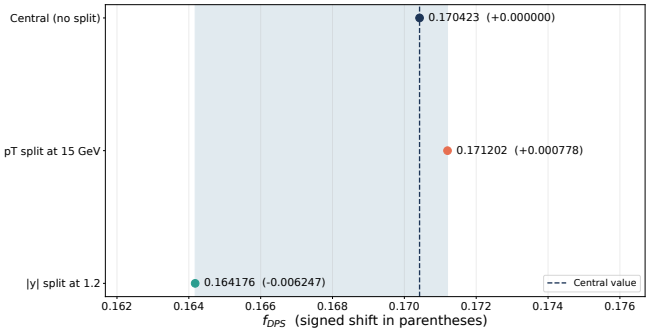


Fixed ensemble: 500/500 successful replicas

Statistical uncertainty

$$\sigma_{\mathrm{stat}}(f_{\mathrm{DPS}}) = \mathrm{SD}(f_{\mathrm{DPS}}^{(b)}) = \mathbf{0.05933}$$

Single-J/ψ category dependence



Category refits

- Split each J/ψ slot independently at $p_{\mathrm{T}} = 15\,\mathrm{GeV}$ or $|y| = 1.2$.
- In every category, repeat the two-stage mass–$c\tau$ fit: calibrate resolutions, then freeze them for the yield-only sPlot.
- Rebuild the mixed DPS template and repeat the full extraction.

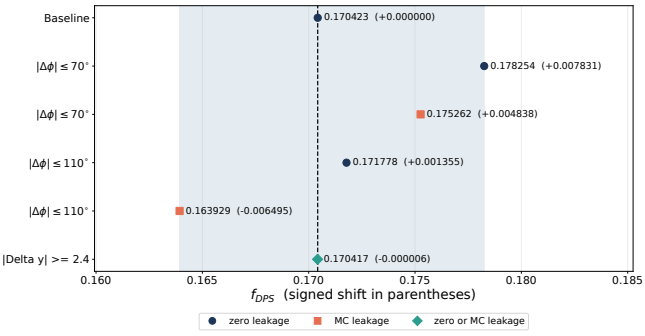
Why this variation?

It probes whether kinematic dependence in the single-J/ψ fit and prompt weights changes the mixed-template shape.

Category envelope

$$\Delta f_{\mathrm{DPS}}^{\mathrm{category}} = {}^{+0.00078}_{-0.00625}$$

Control-region choices set the largest envelope



CR variations

- Change $|\Delta\phi| \leq 90^\circ$ to 70° or 110°; alternatively raise $|\Delta y| \geq 1.8$ to 2.4.
- Refit the PP yields in the new 2D cells, then renormalize the mixed template.

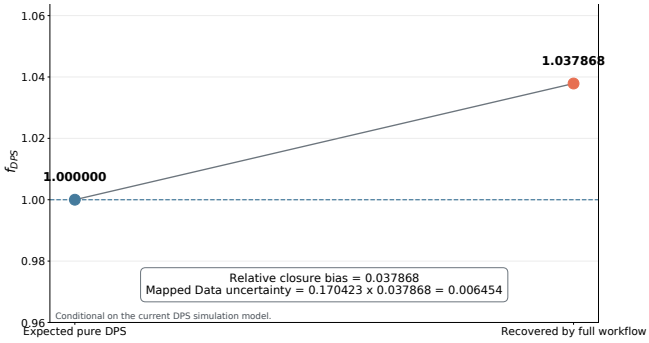
SPS-leakage treatment

- *Zero leakage*: assign the full PP yield in the CR to DPS.
- *MC leakage*: estimate $r_{\mathrm{SPS}}^{\mathrm{CR}}$ from SPS MC and subtract its contribution before setting the DPS normalization.

Envelope over all variants

$$\Delta f_{\mathrm{DPS}}^{\mathrm{CR}} = {}^{+0.00783}_{-0.00649}$$

Pure-DPS closure probes the complete workflow



Where does the sample come from?
A dedicated DPS MC sample supplies same-event J/ψ-pair pseudo-data. It is *not* an event-mixed sample.

What is tested?

- Extract the two single-J/ψ inputs from DPS MC.
- Construct the DPS template by the same cross-event mixing.
- Repeat the 12-cell PP fits, CR normalization and decomposition.

For pure DPS, $f_{\mathrm{DPS}}^{\mathrm{exp}} = 1$; the workflow recovers 1.0379.

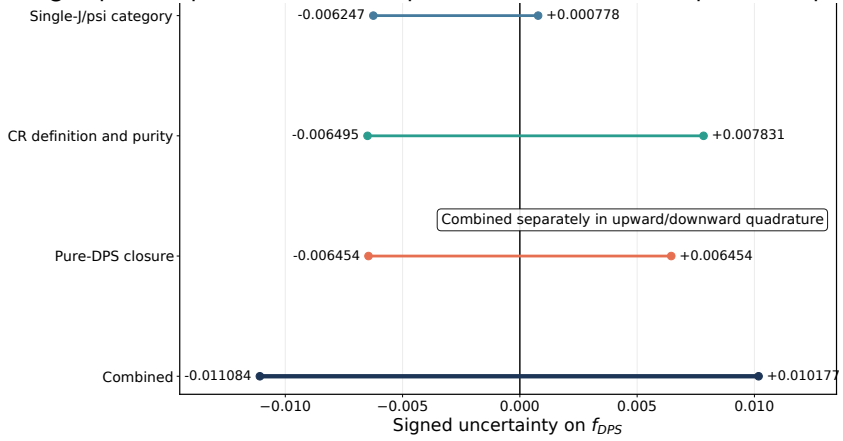
Map the closure bias to Data

$$0.0379 \times 0.1704 = \mathbf{0.00645}$$

symmetric uncertainty

Three method sources combine asymmetrically

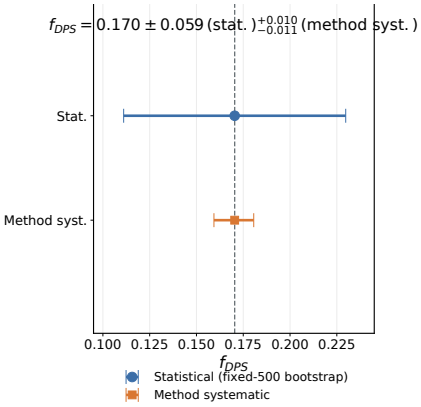
Use each group envelope once; combine upward and downward components separately.

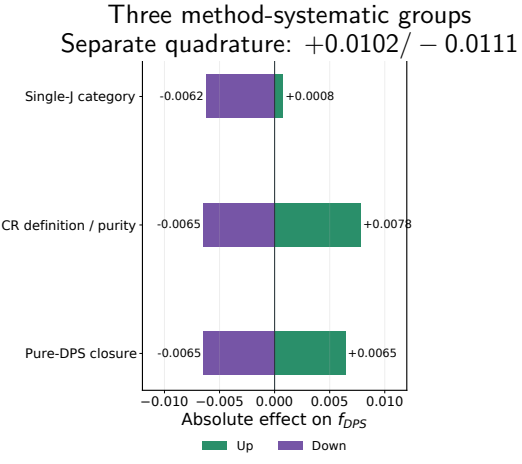


$$\sigma_{\pm} = \sqrt{\sum_{g} \left(\Delta_{g}^{\pm}\right)^{2}}, \quad \Delta f_{\mathrm{DPS}}^{\mathrm{method}} = \begin{matrix} +0.01018 \\ -0.01108 \end{matrix}$$

Final data-driven DPS fraction

The full extraction combines the fixed bootstrap ensemble with three method groups.





The statistical uncertainty dominates the current f_{DPS} precision.

DPS cross section from the data-driven fraction

$$\sigma_{DPS} = f_{DPS}\,\sigma_{pair}$$

$$= 0.170423 \times 37.2 \text{ pb}$$

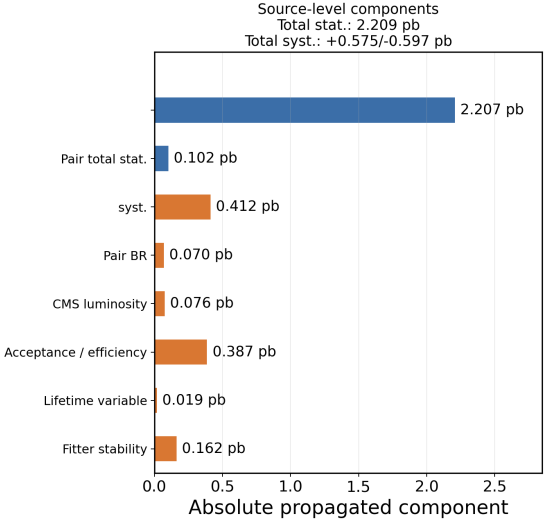
$$6.34 \pm 2.21\,(\text{stat.})^{+0.57}_{-0.60}\,(\text{syst.})\ \text{pb}$$

Fiducial pair region: $m(J/\psi J/\psi) > 7.5$ GeV



For σ_{eff}: divide by $\varepsilon_m = 0.987339$

$$\sigma_{DPS}^{\text{no mass cut}} = 6.421\ \textbf{pb}$$



Effective DPS cross section

$$\sigma_{eff} = \frac{1}{2}\frac{\sigma^2(J/\psi)}{\sigma_{DPS}^{\text{no mass cut}}}$$

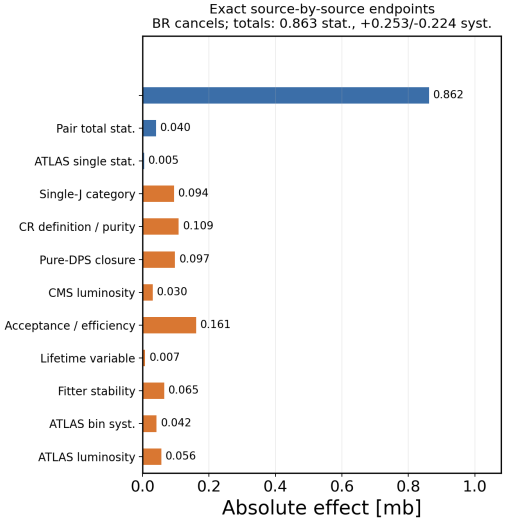
$X = B\,\sigma(J/\psi) = 10.63$ nb

$\varepsilon_m = 0.987339$ from DPS GEN-only event mixing

B cancels between X^2 and the pair-visible denominator

$$\sigma_{eff} = 2.48 \pm 0.86\,(\text{stat.})^{+0.25}_{-0.22}\,(\text{syst.})\ \text{mb}$$

Prompt J/psi: 10 < pT < 40 GeV, |y| < 2.0



The common branching fraction cancels; the pair-mass correction matches the single-J/ψ phase space.

Two separations with complementary inputs

Both methods target the SPS/DPS composition, with different shape information and uncertainty models.

	AN template fit	Data-driven separation
Shape input	SPS/DPS simulation templates	Prompt single-J/ψ data and event mixing
Normalization	Simultaneous differential-cross-section fit	Prompt-prompt yields and control region
f_{DPS}	$0.34 \pm 0.10\,^{+0.02}_{-0.17}$	$0.170 \pm 0.059\,^{+0.010}_{-0.011}$
σ_{DPS} [pb]	$12.6 \pm 3.7\,^{+1.1}_{-6.4}$	$6.34 \pm 2.21\,^{+0.57}_{-0.60}$
σ_{eff} [mb]	$1.26 \pm 0.37\,^{+0.64}_{-0.11}$	$2.48 \pm 0.86\,^{+0.25}_{-0.22}$

Uncertainties are statistical, then systematic.

Conclusions

A complete data-driven SPS/DPS separation is now available for the JJ analysis.

1

Data construction

Prompt-weighted single-J/ψ mixing supplies the DPS shape.

2

Full uncertainty chain

Source-event bootstrap plus three method-systematic groups.

3

Derived observables

σ_{DPS} and σ_{eff} with source-level propagation.

Current precision

Statistical uncertainty is the leading contribution to all three data-driven results.

Thanks for listening!

Reporter: Lekai Yao

29 August 2026